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NepPlural

A Human-Anchored Benchmark for Persona Annotation in Nepali Migration Discourse

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Data, labels, prompts, code

Agreement is not validity. For culturally situated Nepali labels, LLM annotation quality depends on the protocol, and only native-speaker gold can tell good labels from consistent ones.

1 The gap

Alignment that optimises toward one aggregated answer collapses contested views toward a Western-majority default. Pluralistic datasets inherit that skew (PRISM has no participants from Nepal), and Nepali has **no persona or value resource**.

2 NepPlural data

Youth migration, argued online in code-mixed Devanagari, Romanized Nepali and English.

2,148 YouTube comments
3 channels: IDS, Thaha Research, The Nepali Comment; anonymized, every script kept

↓

Rule-based router, no LLM
1,894 substantive · 254 obvious No-Persona

↓

300-comment gold set
Stratified sample of the substantive pool, 3 native speakers

rahar ley haina badhyata ley ho hajur

"It is not by choice but out of compulsion, sir."

Pro · Economic · Individualist · Despairing

3 Four persona axes

Intent
Pro · Anti · Trapped/Regretful · Neutral

Driver
Economic · Family · Systemic Anger · Patriotism

Values
Collectivist-Family · Collectivist-Nation · Individualist-Self

Affect
Despairing · Angry · Hopeful · Pragmatic

+ No-Persona
No migration stance: praise, shout-outs, off-topic remarks

4 Native-speaker gold

38.7%
of substantive-looking comments carry no persona (116 / 300)

0.55
mean Fleiss' κ across 3 native speakers

Humans agree on **whether** there is a persona (κ 0.64) more than **which** (κ 0.42). Against leave-one-annotator-out (LOO) references, a held-out native speaker scores **0.75** macro-F1.

5 Auditing LLM annotation

Same three models (GPT-5.2, Gemini 3.5 Flash, Claude Sonnet 4.6), same instructions, three protocols:

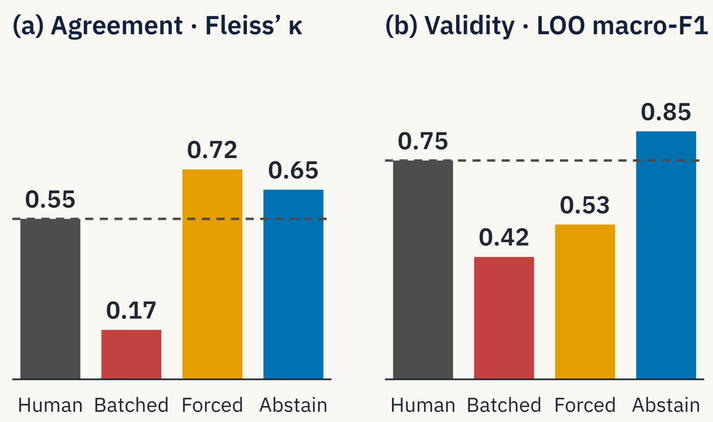
Batched chat
20–50 comments per chat
Forced choice · CSV

Per-comment, forced
One comment per API call
Forced choice · JSON

Per-comment, abstain
One comment per API call
No-Persona allowed · JSON

Forced vs. abstain is a controlled contrast; batched vs. per-comment bundles batching, interface, format and settings.

6 Agreement is not validity



Dashed line: native-speaker level.

Agreement depends on protocol. Batched chat gives near-chance κ 0.17; one comment at a time, the same models reach κ 0.72, above humans.

High agreement can still be wrong. Forced models assign personas to the 39% of comments humans call No-Persona, scoring ≈0.50 macro-F1 vs. gold. On persona-only items they score 0.83–0.88: the failure is over-attribution, invisible to an agreement check.

7 Letting models abstain fixes most of it

0.79–0.87
No-Persona F1

0.75–0.86
LOO macro-F1, vs. 0.75 for a native speaker

Cost: 5–16% false abstentions on persona-bearing comments. Not perfect:

Genuine saving kati samma hunx uta

"How much can one genuinely save over there?"

Humans: No-Persona · all 3 abstaining models: Pro / Economic

8 The labels are the bottleneck

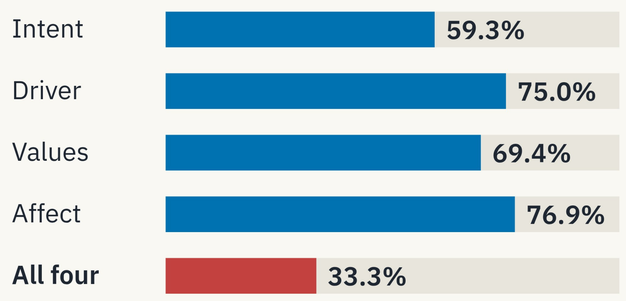
Encoders trained on the batched LLM labels and tested on human gold:

Fine-tuned encoder (5 seeds)	Macro-F1 vs. gold
XLM-RoBERTa base	0.34
mBERT (cased)	0.34
NepBERTa	0.21
IRIIS RoBERTa 125M	0.20
IRIIS BERT 110M	0.20
Batched LLM labels (training data)	**0.39**
Abstaining LLMs, zero-shot	**0.74**

Encoders land near the quality of their training labels. Nepali-specific vs. multilingual gap is confounded with pretraining scale.

9 Steering into personas

6 personas × 6 questions × 3 models = 108 generations, labelled blind by the 3 native speakers. Fidelity to target:



Single traits yes, full personas rarely. No Western-congruence gap detected: value-orientation fidelity 68.5% vs. 70.4% (gap –1.9 pts, 95% CI [–20.4, +16.7]). This rules out only gaps of about 20 points or more.

10 Takeaways

1. **Report and audit the protocol:** reliability is a property of how annotation is run.
2. **Anchor on native speakers:** agreement is not validity for culturally situated labels.
3. **Let models abstain:** knowing when not to speak for someone is a capability.

Limitations. One domain and platform; 300-item gold with small script groups; annotators on the research team; proprietary models only; small steerability study.

github.com/iriis-research/NepPlural
arXiv 2610.15732